

EUROPEAN ECONOMICS ANALYST

Rockets and Feathers: Asymmetric Passthrough from Energy Shocks

- Studies show that energy price shocks tend to pass through to underlying inflation asymmetrically: rising energy prices push core inflation up much more than falling prices pull it down. This “rocket and feather” pattern matters for the ECB because it implies that renewed increases in energy prices could create persistent inflation pressure, while a decline in energy prices may offer only limited relief.
- Research highlights three main reasons for this asymmetry: firms protect margins more aggressively when costs rise than when they fall (given the possibility of bankruptcy), the “menu” cost of lowering prices tends to be higher with positive trend inflation, and wages and prices are sticky on the downside. Together, these mechanisms help explain why inflation can become more persistent after an energy spike than after an equivalent decline.
- We provide fresh statistical evidence on this uneven passthrough behaviour in the Euro area. We start with a simple top-down perspective using quarterly data from 1999–2025. We confirm that positive food and energy price shocks have significant passthrough effects into core inflation, while the passthrough of negative non-core shocks is small and insignificant.
- We show that this finding is robust to considering a broader set of data. First, we estimate a panel version of the top-down model, which utilises the cross-country variation in energy shocks to provide more statistical information. Second, we create a sectoral panel model of firm pricing intentions using the European Commission business survey.
- For the outlook, asymmetric passthrough skews risks to the upside for core inflation because further energy price increases would pass through more forcefully than declines. Uneven passthrough could also raise the chance of ECB tightening by lowering the odds that a sudden drop in energy prices improves the inflation outlook materially and by strengthening the case for risk management hikes. Our analysis thus supports our recent upgrade to core inflation (which we now see peaking at 2.7%) and our baseline forecast for two ECB hikes (June and September).

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Rockets and Feathers: Asymmetric Passthrough from Energy Shocks

ECB officials are focused on passthrough effects from higher energy prices into underlying inflation, via indirect effects (through supply chains) and second-round effects (through wage bargaining). Our previous [work](#) shows that the likelihood of such passthrough depends importantly on the size of the shock, with larger energy price increases more likely to lead to persistent inflationary pressures.

Past energy shocks also point to a potential asymmetry in the transmission of energy shocks. In particular, energy price *increases* tend to have stronger effects for underlying inflation than energy price *declines*. This asymmetry is often called “rocket and feather” pricing.

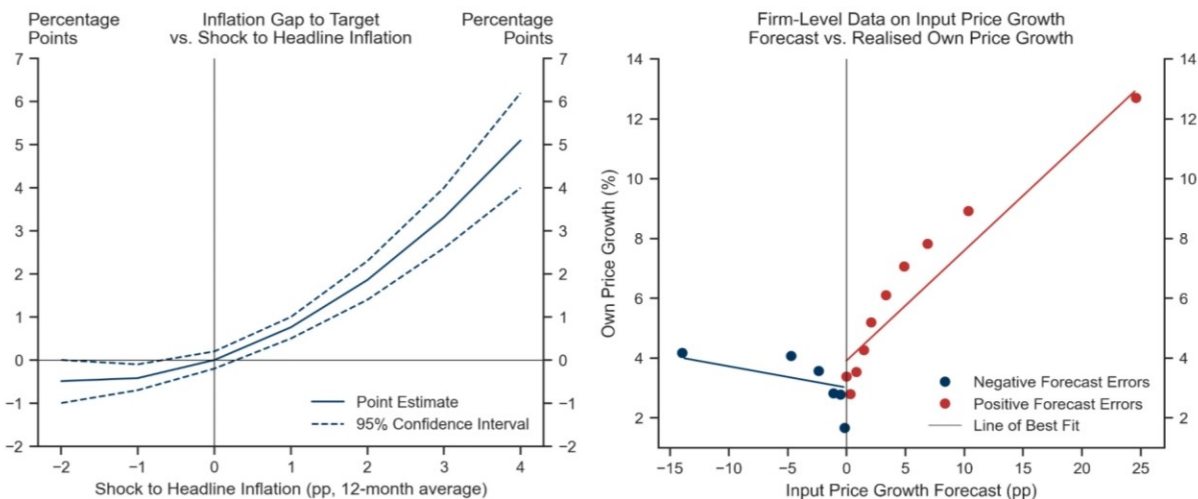
Given the ongoing uncertainty around energy prices, such asymmetries in passthrough could have important implications for the inflation and policy outlook.

Existing Evidence on Asymmetric Passthrough

Several studies point to asymmetric transmission from energy prices to inflation.

The IMF, for example, has [documented](#) uneven passthrough from headline inflation shocks to core inflation in the Euro area, with positive food and energy inflation shocks displaying a strong increasing relation with core inflation, whereas negative headline shocks have an insignificant relationship with core inflation (Exhibit 1, left).

Exhibit 1: Previous Studies Point to Uneven Passthrough of Energy Shocks



On the left, the chart is visually approximated from Dao et al. (2023). On the right, the chart is visually approximated from De Fiore et al. (2025).

Research from Denmark’s Nationalbank similarly [found](#) that higher natural gas prices have a significantly greater and more persistent impact on Euro area inflation, whereas a negative shock of the same magnitude had an almost negligible effect. For crude oil, the asymmetry was also present but less pronounced.

Similar insights emerge from micro data at the firm level. For example, a [study](#) of the French manufacturing sector estimated a passthrough rate for positive energy price shocks of over 100%, compared to just 58% for negative shocks. A BIS [study](#) similarly found that Italian corporate pricing responds strongly to rising input costs but only

weakly to falling ones (Exhibit 1, right).

Why the Asymmetry?

Uneven passthrough could reflect several factors.

First, asymmetric firm markup adjustment. When input costs rise, firms are forced to adjust prices upward to protect their profit margins given the constraint of bankruptcy. But when costs fall, firms might retain prices to expand margins, especially when they have market power.

Second, the cost of changing prices (“menu costs”) could play a role. Given the cost of changing prices—both physical “menu costs” and the desire to avoid negative reactions to frequent price changes—firms may absorb small cost decreases rather than incur these broader adjustment costs with positive trend inflation.

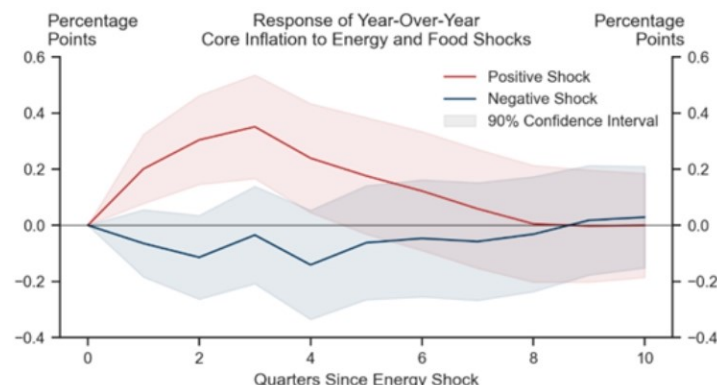
Third, the asymmetry could reflect downward nominal stickiness in prices and wages. When a positive energy shock erodes purchasing power, workers are likely to demand higher wages. Conversely, firms find it very difficult to cut nominal wages when their costs fall, meaning cost savings are not fully passed on through lower labour costs.

New Evidence

We provide fresh statistical evidence on this passthrough behaviour in the Euro area.

We start with a simple top-down perspective, similar to the IMF and Danish national central bank. In particular, we use a statistical (vector autoregressive) model to estimate the effect of non-core (food and energy) inflation on subsequent core inflation, controlling for long-term inflation expectations and labour market slack. Using a quarterly sample from 1999–2025 we find that commodity price shocks have statistically significant effects on subsequent core inflation. This passthrough, however, is entirely driven by positive price shocks, while the passthrough of negative non-core shocks is small and insignificant (Exhibit 2).

Exhibit 2: Top-Down Evidence Clearly Shows an Asymmetry in Food and Energy Shock Passthrough



Source: Goldman Sachs Global Investment Research

This result, however, is sensitive to the inclusion of the 2022–23 energy shock. Restricting the sample to 1999–2019 reveals no significant evidence for asymmetric passthrough behaviour.

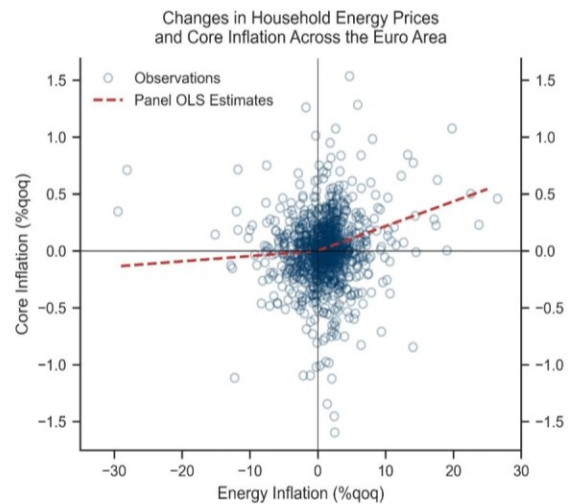
We therefore consider two extensions that consider a broader set of data.

The first is a panel version of the model above, which utilises the cross-country variation in energy shocks and passthrough to provide more statistical information. We find that passthrough is significant for positive energy inflation, but statistically insignificant for negative energy price growth (second columns in Exhibit 3, left). In other words, the relationship between energy inflation and core inflation appears to be “kinked” (as shown in Exhibit 3, right). Our results are also robust to ending the sample in 2019 (third column in Exhibit 3, left).

Exhibit 3: A Cross-Country Panel Confirms Uneven Passthrough

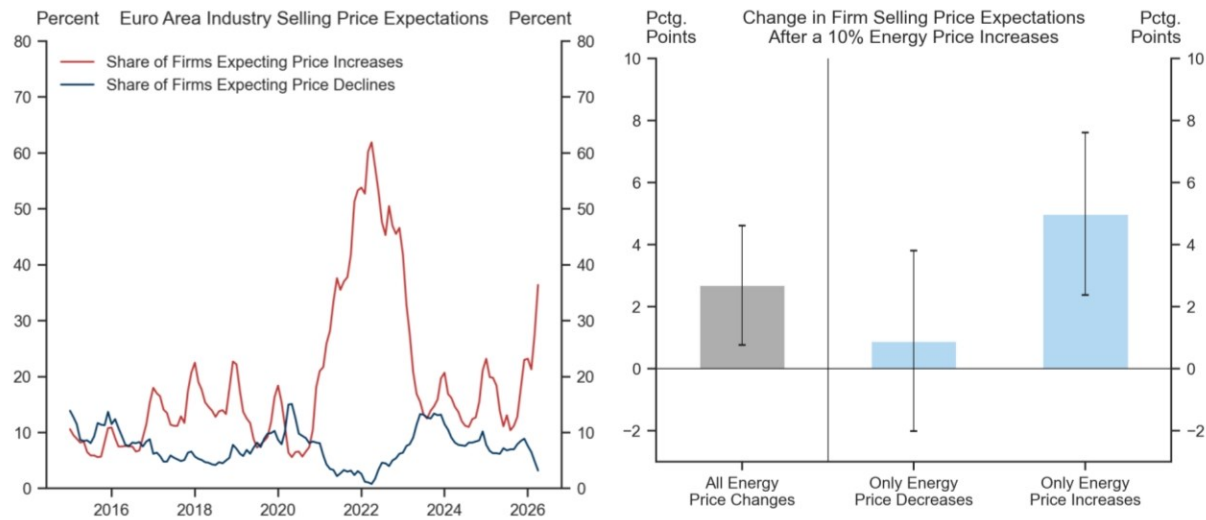
	Dependent variable: Core Inflation (%qoq, non-ann.)		
	(1) Linear 2000-2025	(2) Kinked 2000-2025	(3) Kinked 2000-2019
Core Inflation (%qoq, non-ann., 1q lag)	0.17***	0.16***	0.11**
Core Inflation (%qoq, non-ann.d, 2q lag)	0.23***	0.23***	0.29***
Inflation Expectations (%qoq, non-ann.)	0.48***	0.47***	0.44***
Labour Market Tightness (standard deviations, 1q lag)	0.12***	0.12***	0.12***
Energy Inflation (%qoq, non-ann., 1q lag)	0.01***		
Positive Energy Inflation (%qoq, non-ann., 1q lag)		0.02***	0.02***
Negative Energy Inflation (%qoq, non-ann., 1q lag)		0.00	0.00
Fixed Effects	Country	Country	Country
Observations	1027	1027	798
R-squared	0.49	0.49	0.37

*, ** and *** means significance at 10/5/1% levels under Driscoll and Kraay (1998) standard errors.



Source: Goldman Sachs Global Investment Research

The second is a panel of firm pricing intentions, using the European Commission business survey. The intuition is that firms look much more likely to report selling price increases after energy prices rise—such as in 2021 and 2022—than report selling price decreases when energy prices fall—like in 2023 and 2024 (Exhibit 4, left). We test for this statistically by regressing selling price expectations across sectors on the product of wholesale energy prices and each sector’s energy intensity, as an “instrument” for sector-level energy prices. We also control for employment expectations to capture aggregate demand.

Exhibit 4: Uneven Pricing is also Found in Sectoral Firm Pricing Surveys

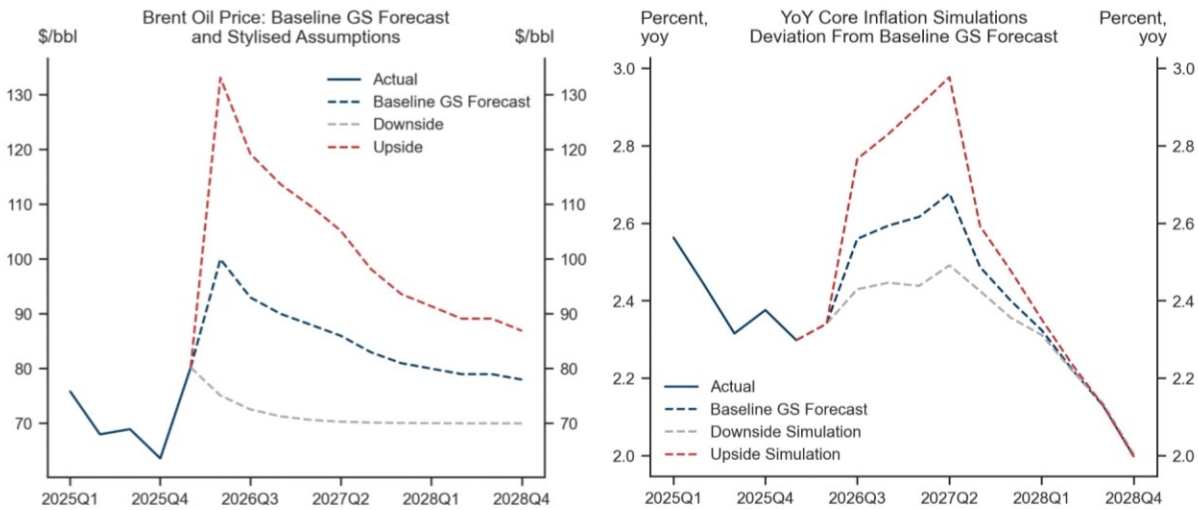
Source: Goldman Sachs Global Investment Research, European Commission

Our results suggest that firms in more energy-intensive sectors are more likely to report higher selling price expectations when energy prices increase (Exhibit 4, right). In contrast, a fall in energy prices does not lead to a statistically significant change in firms' selling price expectations. This confirms that firms are more likely to adjust prices after a rise than a fall in energy prices.

Implications for the Outlook

We see two implications of this “rocket and feather” passthrough for the outlook.

First, the asymmetric passthrough creates upside risks to the core inflation outlook. That is, declines in energy prices from here (following a reopening of the Strait of Hormuz) might provide little relief for the core inflation outlook, while further energy price increases could push up inflation sharply further. This can be seen by mapping symmetric energy price assumptions into core inflation using our cross-country panel regression results (Exhibit 5).

Exhibit 5: The Pricing Asymmetry Creates Upside Risks for Core Inflation

On the left, the downside assumption converges towards 70\$/bbl. The upside assumption features the same but positive deviation from the baseline, so as to be symmetric to the downside assumption.

Source: Goldman Sachs Global Investment Research, Haver Analytics

Second, uneven passthrough could support the case for ECB rate hikes. This is both by lowering the chance that a sudden drop in energy prices could improve the inflation outlook significantly and by strengthening the case for risk management hikes in the presence of upside inflation risks.

Taken together, our analysis supports our recent upgrade to the core inflation profile (which we now see peaking at 2.7%) and our baseline forecast for two ECB hikes (June and September).

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